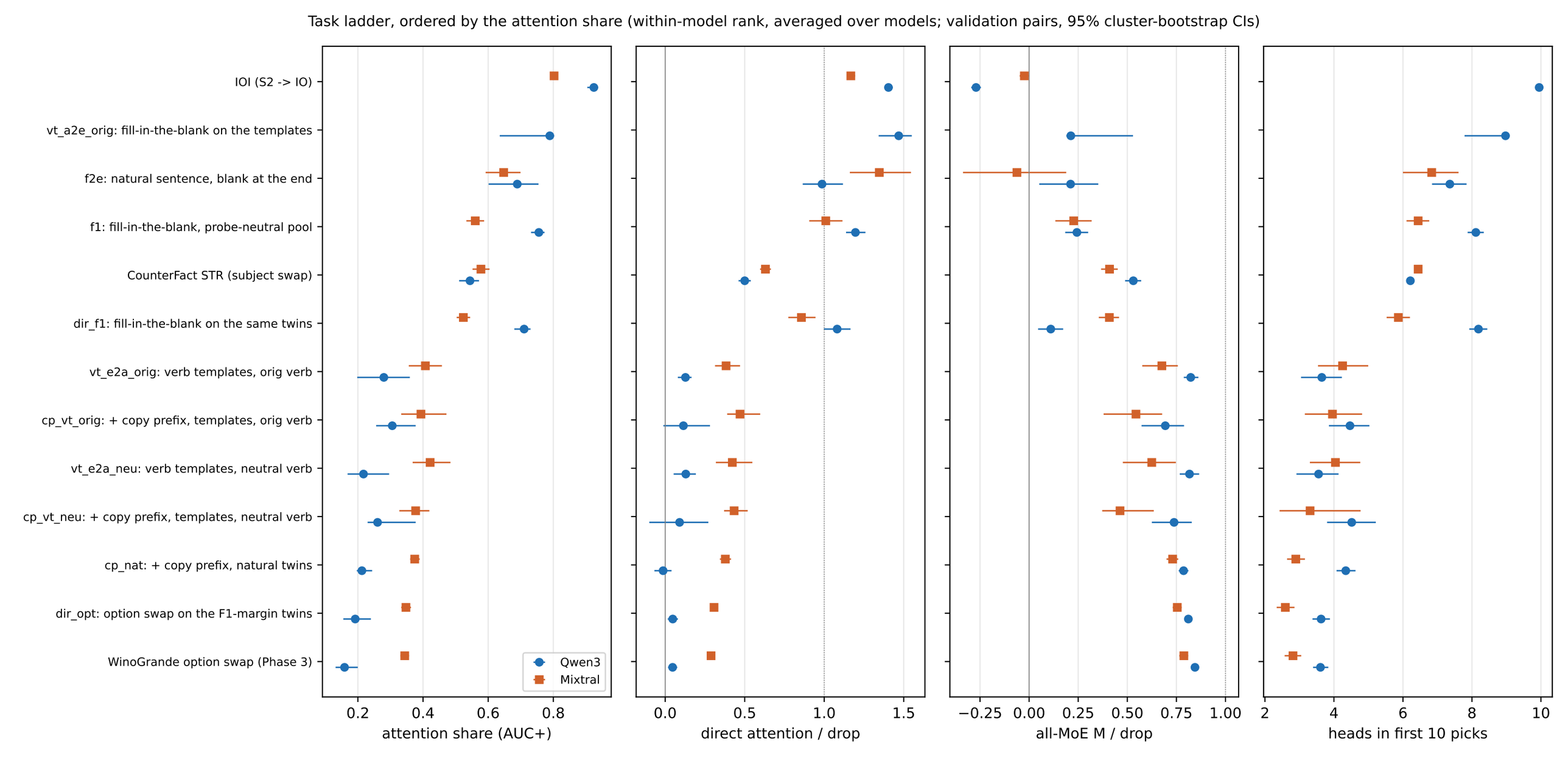
Task ladder, ordered by the attention share (within-model rank, averaged over models; validation pairs, 95% cluster-bootstrap CIs)
IOI (S2 -> IO)
vt_a2e_orig: fill-in-the-blank on the templates
f2e: natural sentence, blank at the end
f1: fill-in-the-blank, probe-neutral pool
CounterFact STR (subject swap)
dir_f1: fill-in-the-blank on the same twins
vt_e2a_orig: verb templates, orig verb
cp_vt_orig: + copy prefix, templates, orig verb
vt_e2a_neu: verb templates, neutral verb
cp_vt_neu: + copy prefix, templates, neutral verb
cp_nat: + copy prefix, natural twins
dir_opt: option swap on the F1-margin twins
WinoGrande option swap (Phase 3)
Qwen3
Mixtral
0.2
0.4
0.6
0.8
attention share (AUC+)
0.0
0.5
1.0
1.5
direct attention / drop
−0.25
0.00
0.25
0.50
0.75
1.00
all-MoE M / drop
2
4
6
8
10
heads in first 10 picks